

VELS

INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
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PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR



FRESHAIR DESK

AI- Powered Smart Desk Environment Monitor

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PROJECT GUIDE:

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INTRODUCTION

- People spend a significant amount of time in indoor environments such as homes, offices, and classrooms.
- Indoor environmental conditions can directly affect comfort, productivity, and well-being.
- However, these conditions are often not monitored continuously in a simple and integrated manner.
- FreshAir Desk is a smart indoor environment monitoring system that measures temperature, humidity, air quality, and light intensity using multiple sensors.
- An ESP32 is used for data acquisition and processing to assess the overall environmental condition.
- The system aims to provide real-time, compact, and low-cost environmental monitoring for a workspace

ABSTRACT

- FreshAir Desk is an IoT-based smart device for monitoring indoor desk environments.
- Uses ESP32 with sensors to measure temperature, humidity, light intensity, and air quality.
- Collects and processes environmental data in real time.
- Analyzes the data to generate Comfort, Focus, and Air Quality Scores.
- Provides recommendations to improve the surrounding environment.
- Combines IoT, Embedded Systems, Environmental Sensing, and AI for smart environmental monitoring.

BASE PAPER: Contributions, Methodology, Dataset, Limitations

- Reference: Bakare & Abubaker (2026), Discover Electronics — MCIMS system
- Contribution: reproducible, calibrated multi-sensor IoT platform with MQTT integration
- Methodology: ESP32-WROOM with DHT22, BH1750, and DSM501A sensors
- Dataset: 3 indoor sites at KIU, 3-day deployment, 90 samples/node
- Achieved $R^2 > 0.98$, RMSE < 0.5, and 99.1% packet delivery
- Limitation: no personalized feedback; short deployment; no CO_2/VOC sensing

LITERATURE SURVEY

FreshAir Desk – AI-Powered Smart Desk Environment Monitor

Author & Title	Methodology	Limitations	Our Contribution to Overcome the Limitations
S. Shriram, V. Rajesh Kumar, et al. "Towards Healthier Workspaces: IoT-Based Thermal Comfort and Indoor Air Quality Monitoring" (ICCoSD, 2025)	IoT-based monitoring of temperature, humidity and air quality for workplace/classroom comfort.	No AI-based prediction; limited environmental parameters; no personalized recommendations.	Integrates multiple environmental sensors with AI-based scoring and personalized suggestions for comfort, focus and air quality.
T. Praveenraja, P. V. Manitha "Integration of IoT-Based Air and Sound Pollution Monitoring with HVAC System for Enhanced Indoor Environment Management" (ICOECA, 2025)	IoT monitoring of air quality and noise with HVAC integration for automatic environmental control.	Higher implementation cost; requires HVAC infrastructure; limited AI capabilities.	Uses a low-cost ESP32-based desk system with AI-based recommendations, avoiding dependence on full HVAC infrastructure.
M. Gabriel, T. Auer "Indoor Air Pollution Estimation Using Machine Learning" (BauSim, 2022)	ANN and SVR models estimate indoor air pollution from environmental sensor data.	Requires large training dataset; high computational complexity; focuses only on air quality.	Extends monitoring beyond air quality to temperature, humidity, light and noise, with lightweight environment/comfort-oriented decision support.
D. Yeom, T. Kim, S.-G. Yoon "Productivity Prediction via Human Physiological Signals for an Optimum Thermal Environment" (CLIMA, 2022)	AI model predicts human productivity using thermal conditions and physiological signals.	Requires wearable sensors; complex implementation; small experimental dataset.	Provides desk-level focus/comfort assessment using environmental sensing without requiring wearable physiological sensors.

LITERATURE SURVEY

FreshAir Desk – AI-Powered Smart Desk Environment Monitor

Author & Title	Methodology	Limitations	Our Contribution to Overcome the Limitations
G. Cicceri, et al. "Smart Healthy Intelligent Room: Headcount through Air Quality Monitoring" (IEEE SMARTCOMP, 2020)	Machine learning and air-quality sensors estimate room occupancy for smart-building monitoring.	Limited environmental monitoring; mainly focused on occupancy; no comfort or productivity scoring.	Focuses on the user's desk environment and adds Comfort Score, Focus Score and Air Quality Score with actionable recommendations.
F. Fanti, et al. "An IoT-Based Smart Building Solution for Indoor Environment Management and Occupants Prediction" (Energies, 2021)	IoT-enabled smart-building system using machine learning for environmental management and occupancy prediction.	Complex architecture; high deployment cost; requires large datasets.	Targets a compact, low-cost desk-scale implementation using ESP32 and multiple sensors with an AI recommendation layer.
Multiple Authors "Comparative Analysis of Indoor Environmental Quality and Self-Reported Productivity in Intelligent and Traditional Buildings" (Energies, 2023)	Compares indoor environmental quality and self-reported productivity in intelligent and traditional buildings.	Based on user surveys; limited AI implementation; no real-time monitoring system.	Adds real-time sensor monitoring and AI-based scoring/recommendations to connect environmental conditions with workspace focus and comfort.
M. S. Bakare, K. Abubaker "IoT-Based Indoor Environmental Monitoring System Using Multi-Parameter Sensing and ESP32-WROOM Integration" (Discover Electronics, 2026)	Low-cost ESP32 system measuring temperature, humidity, light intensity and particulate matter in real time.	No intelligent recommendation system; limited predictive analysis; mainly focused on monitoring.	Builds on multi-parameter ESP32 monitoring by adding AI-based environment classification/scoring and personalized recommendations.

FreshAir Desk

AI-Powered Smart Desk Environment Monitor | System Architecture

Sensors

(Environmental Data Collection)

- **DHT22** — Temperature & Humidity
- **BH1750** — Light Intensity (Lux)
- **MQ135** — Air Quality (Gas/Smoke)
- **LED Indicators** — Status Indication
- **Resistors** — Current Limiting / Pull-up / Pull-down

Sensor Data (Analog/Digital)

ESP32

(Data Processing & Web Server)

- Reads sensor values
- Processes & validates data
- Connects to Wi-Fi
- Hosts web server
- Provides data via JSON API

Wi-Fi (Local Network)

User Dashboard

(Web Interface)



- Displays real-time sensor data
- Shows historical data (30s)
- Auto refresh (5 seconds)
- Shows overall status & recommendations
- Fetches data via /json endpoint

AI Recommendation

AI Recommendation Layer (Future Enhancement)

Sensor Data (from Dashboard)

→

Python Backend (API Middleware)

- Receives sensor data
- Calls OpenAI API
- Gets recommendation

→

OpenAI API (Generates AI Insights)

- Analyzes combined data
- Creates natural language advice
- Returns recommendation

→

AI Recommendation (Displayed on Dashboard)

"The room is warm and humid. Improve ventilation and consider using a desk lamp."

Key Features

- Real-time Monitoring
- Historical Data
- Wi-Fi Connectivity (IoT)
- Error Handling & Validation
- AI-Powered Recommendations
- User-Friendly Dashboard

FreshAir Desk: Module Flow

Sensors → ESP32 rules → live dashboard → laptop AI server → AI recommendations

MODULE 1

Environmental Sensors

DHT22
Temperature
Rel. humidity

BH1750
Ambient light
Lux

MQ135
Gas response
RS/R0 ratio

Sensors send readings to the ESP32

MODULE 2

ESP32 + Rule-Based Status

Reads sensors repeatedly
Compares values with rules
Sets Good / Attention / Poor
Output: red / green LEDs

Prototype thresholds, not a certified rating

MODULE 3

Live Dashboard

ESP32 serves readings at /json
Browser shows temp, humidity,
light and MQ135 RS/R0
Status labels, refresh every 5 s

Dashboard runs locally on the network

latest readings (/json)

advice shown beside readings

EXTERNAL SERVICE

Groq API

Hosted language model
Creates recommendation text
from the readings sent to it

MODULE 4

Laptop AI Server

Python + Flask service
Fetches latest readings
Builds AI request, returns response

API key stored privately in .env

MODULE 5

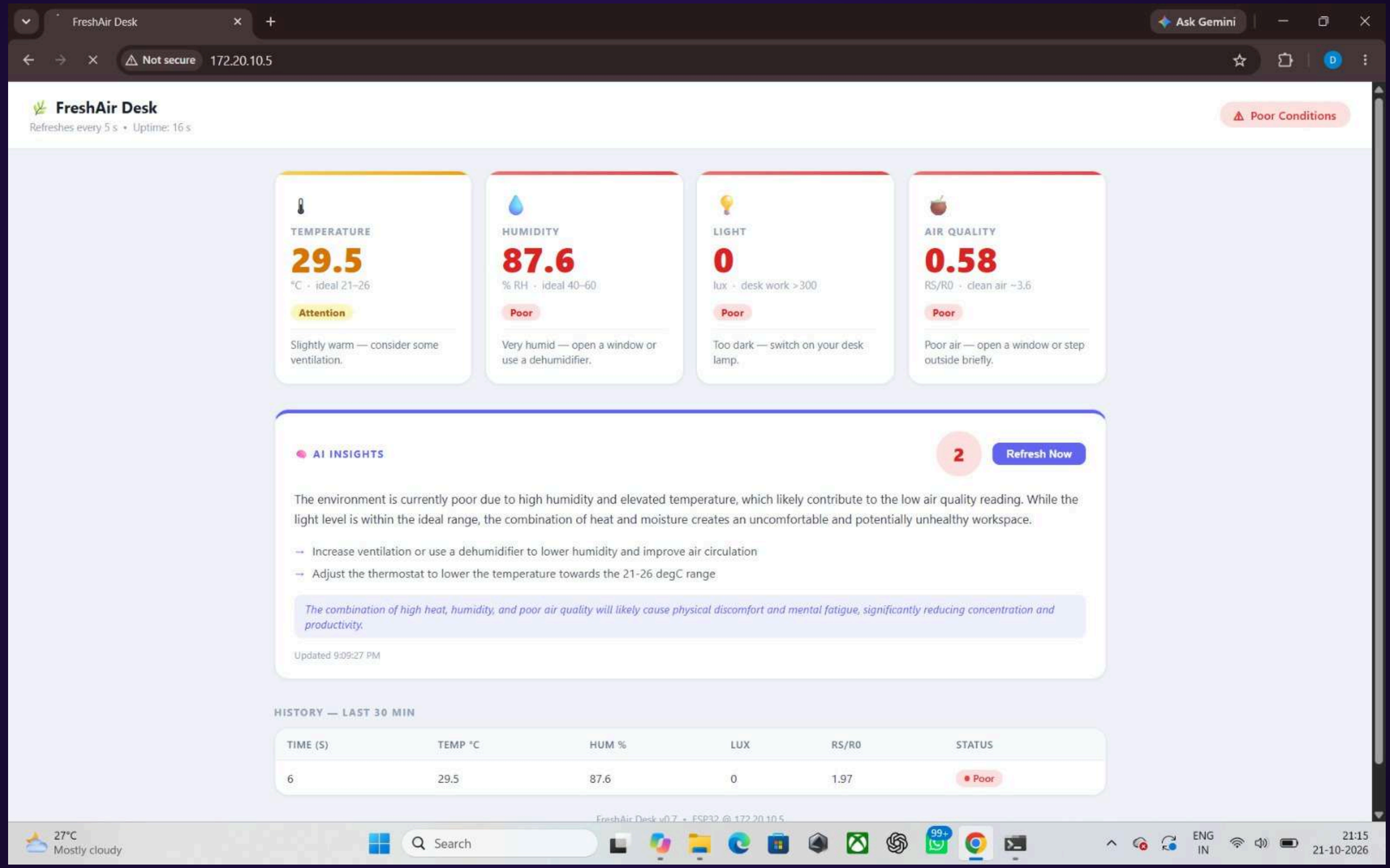
AI Recommendations

Summarizes conditions
Suggests possible actions
Needs human review

Advisory only: no automatic appliance control

Suggestions are advisory. The ESP32 does not run the AI model. Sensor dashboard works locally; AI insights need the laptop service and internet.

RESULT


FreshAir Desk
Ask Gemini
Not secure
172.20.10.5
FreshAir Desk
Refreshes every 5 s • Uptime: 16 s
Poor Conditions
TEMPERATURE
29.5
°C · ideal 21–26
Attention
Slightly warm — consider some ventilation.
HUMIDITY
87.6
% RH · ideal 40–60
Poor
Very humid — open a window or use a dehumidifier.
LIGHT
0
lux · desk work >300
Poor
Too dark — switch on your desk lamp.
AIR QUALITY
0.58
RS/R0 · clean air ~3.6
Poor
Poor air — open a window or step outside briefly.
AI INSIGHTS
2
Refresh Now
The environment is currently poor due to high humidity and elevated temperature, which likely contribute to the low air quality reading. While the light level is within the ideal range, the combination of heat and moisture creates an uncomfortable and potentially unhealthy workspace.
→ Increase ventilation or use a dehumidifier to lower humidity and improve air circulation
→ Adjust the thermostat to lower the temperature towards the 21-26 degC range
The combination of high heat, humidity, and poor air quality will likely cause physical discomfort and mental fatigue, significantly reducing concentration and productivity.
Updated 9:09:27 PM
HISTORY — LAST 30 MIN
TIME (S) | TEMP °C | HUM % | LUX | RS/R0 | STATUS
6 | 29.5 | 87.6 | 0 | 1.97 | Poor
FreshAir Desk v0.7 • ESP32 @ 172.20.10.5
27°C
Mostly cloudy
Search
ENG
IN
21:15
21-10-2026

REFERENCES

[1] S. Shriram, V. Rajesh Kumar, et al., “Towards Healthier Workspaces: IoT-Based Thermal Comfort and Indoor Air Quality Monitoring,” International Conference on Computing, Communication and Sustainable Development (ICCoSD), 2025.

[2] T. Praveenraja and P. V. Manitha, “Integration of IoT-Based Air and Sound Pollution Monitoring with HVAC System for Enhanced Indoor Environment Management,” International Conference on Electronics, Computing and Communication Applications (ICOECA), 2025.

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[10] A. E. Afful, J. Ayarkwa, G. K. K. Acquah, and D. Osei-Asibey, “Enablers of Incorporating Indoor Environmental Quality (IEQ) Principles into Buildings,” Smart and Sustainable Built Environment, vol. 12, no. 1, pp. 38–59, 2023.

THANK YOU SO MUCH